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Original Research

Selection-based prompting with synthesized candidates unlocks high-performance clinical abbreviation disambiguation via open-source LLMs

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ABSTRACT

Objective: Clinical abbreviations introduce semantic ambiguity that hinders automated understanding in healthcare informatics. While generative large language models (LLMs) show promise, direct generation often lacks clinical precision. We present S-ACAD, a task-specific framework that shifts the paradigm from generation to constrained selection using dual-pathway candidate construction to improve clinical abbreviation disambiguation.

Methods: The S-ACAD pipeline consists of four stages: (i) Anchor generation via preliminary LLM expansion; (ii) Phrase-level retrieval of the top-3 semantic candidates; (iii) Contextual prototyping using pseudo-texts to identify three additional candidates; and (iv) Discriminative selection from the resulting 7-option pool. We evaluated the framework on the original and Adams's denoised CASI datasets using six open-source LLMs.

Results: S-ACAD demonstrates superior performance, with Gemma-2-9B-IT emerging as the best-performing model, achieving remarkable results on Adams's denoised CASI dataset (accuracy: 0.9228, Macro-F1: 0.9279). Even on the challenging original dataset, it maintains high consistency (accuracy: 0.7585, Macro-F1: 0.7585). Ablation studies confirm that shifting the paradigm from generation to selection via dual-pathway candidate construction is the primary driver of performance, significantly outperforming direct generative baselines.

Conclusion: S-ACAD reframes clinical abbreviation disambiguation as a constrained selection task for generative LLMs, effectively unlocking the reasoning potential of small models through dual-pathway candidate construction. This offers a high-precision, economical, and privacy-preserving solution for standardizing electronic health records, thereby enhancing their downstream utility in clinical informatics.

1. Introduction

The widespread adoption of Electronic Health Record (EHR) systems and the growing imperative to reuse historical clinical data highlight the significance of extracting structured information from unstructured clinical texts, such as medical notes and diagnostic reports [1]. However, this task is complicated by the frequent ambiguity of abbreviations in clinical narratives [2]. Such abbreviations can denote specialized medical terms, general language concepts, or domain-specific shorthand. For instance, "mg" may contextually refer to either "myasthenia gravis" or "milligram" [3]. Moreover, standard medical terminologies often provide only full concept names, omitting their common abbreviated forms. For example, LOINC [4] lists the laboratory test "Hemoglobin A1c [Mass/volume] in Blood" (code 41995-2) using its official name, without including widely used abbreviations like "HbA1c". Furthermore, the training and fine-tuning of Large Language Models (LLMs) critically rely on semantically well-defined textual data [5]. Ambiguous abbreviations can cause semantic inconsistencies, impairing model comprehension and generative abilities. Therefore, automated abbreviation disambiguation is crucial for improving clinical text readability, enhancing Natural Language Processing (NLP) task performance in the clinical domain, and standardizing clinical data. It also aids in creating higher-quality training corpora for LLMs.

Clinical abbreviation disambiguation methods include rule-based

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[6,7], machine learning [2,4,8–13], deep learning [14–18], and LLM-driven strategies [19,20]. Contextual understanding is vital, but rule-based string matching fails to capture it fully. In contrast, supervised machine and deep learning methods are limited by their heavy reliance on large-scale, abbreviation-specific annotations. For instance, while fine-tuning models like BioClinicalBERT can achieve high accuracy [21], such performance depends on substantial annotation resources and complex data processing workflows.

Recently, LLMs offer novel opportunities for abbreviation disambiguation, largely owing to their exceptional contextual comprehension capabilities [22–25]. Related studies also confirm the powerful performance of LLMs in this domain. For instance, Hosseini et al. [26] achieved a high accuracy on the denoised CASI dataset by combining the BERT model with BioGPT-based data augmentation. Kugic et al. [19] demonstrated ChatGPT's proficiency in understanding complex, multilingual clinical contexts, highlighting its potential to enhance disambiguation accuracy. Similarly, Agrawal et al. [27] demonstrated that LLMs like GPT-3 can achieve strong zero-shot performance, reaching 0.8600 accuracy on Adams's denoised CASI dataset, thereby establishing a robust benchmark. On this basis, Kugic et al. [20] further improved the accuracy by applying GPT-4 with contextual clues. Despite the high accuracy reported by proprietary models like GPT-4, their application in clinical environments is severely constrained by two primary factors. First, the reliance on cloud-based APIs introduces significant risks to patient data privacy and incurs prohibitive operational costs [28]. For healthcare institutions, the requirement for absolute data sovereignty often renders cloud-dependent solutions non-viable. Therefore, we argue that any potential performance gap between open-source models and GPT-4 is a necessary and justifiable trade-off for ensuring a locally deployable, privacy-preserving infrastructure. Second, the inherent lack of controllability in generative LLMs frequently leads to 'hallucinations' [29] and expansions that deviate from standardized medical terminologies [30]. Evidence suggests that the bottleneck in clinical abbreviation disambiguation is not a lack of internal medical knowledge within LLMs, but rather the instability of the open-ended generative form, which fails to consistently align with structured clinical inventories.

To address these challenges, this study proposes S-ACAD, a novel **S**election-anchored framework for **A**utomatic **C**linical **A**bbreviation **D**isambiguation that shifts the task from open-ended generation to discriminative selection for LLMs. We hypothesize that the primary bottleneck in LLM-based disambiguation is not a lack of internal clinical knowledge, but rather the instability of the open-ended generative format. By transforming the task into a structured multiple-choice format using dual-pathway candidate construction, S-ACAD can more effectively leverage the superior reasoning and discriminative capabilities of open-source LLMs to achieve precise and standardized outputs.

Specifically, this methodology involves an initial query to the LLM to capture a preliminary semantic interpretation of the abbreviation, serving as a generative anchor. Then, a candidate pool is constructed through dual-pathway retrieval: (1) semantic alignment with clinical abbreviation sense inventories using the anchor, and (2) contextual matching using LLM-generated pseudo-text prototypes. Instead of relying on a single generative output, our method synthesizes these diverse evidence sources into a 7-option candidate pool. The LLM is then tasked with acting as a clinical judge, selecting the most appropriate expansion from this structured list.

Our experimental results, particularly on Adams's denoised CASI dataset, demonstrate that this selection-anchored approach yields a dramatic performance leap. By providing a structural safeguard through curated options, even lightweight open-source models (with fewer than 9B parameters) can achieve high accuracy, significantly outperforming their direct generative baselines. This study reveals the untapped potential of open-source LLMs in clinical NLP and offers a practical, scalable, and high-precision solution for standardizing medical text.

The contributions of the study are threefold:

- A task-specific framework for clinical abbreviation disambiguation: We propose S-ACAD, a framework that reformulates clinical abbreviation disambiguation as a discriminative selection task rather than an open-ended generation problem. The framework utilizes anchor-guided semantic retrieval and synthetic context prototypes to facilitate a dual-pathway candidate construction process, ensuring terminological precision in clinical contexts.
- Absolute data sovereignty and privacy-centric deployment: This study provides a cloud-independent solution that allows hospitals to maintain absolute data sovereignty. By successfully leveraging open-source models, S-ACAD ensures that sensitive patient information remains strictly within the local infrastructure, addressing the critical privacy, security, and ethical barriers that currently prevent the adoption of proprietary, cloud-based APIs like GPT-4 in healthcare settings.
- Performance optimization for resource-constrained environments: Our framework proves that small-scale models (<9B parameters) can achieve results comparable to much larger proprietary models through optimized task framing. This democratizes high-performance clinical NLP, enabling medical institutions to deploy state-of-the-art tools on standard local hardware without the prohibitive costs of massive GPU clusters or expensive subscription fees.

Statement of significance

Heading	Summary
Problem or Issue	Clinical abbreviations are highly ambiguous, leading to errors in automated EHR processing and hindering the secondary use of healthcare data.
What is Already Known	Generative LLMs show promise but often lack precision and "hallucinate" expansions because they rely on open-ended generation without structured constraints.
What this Paper Adds	We introduce **S-ACAD**, a task-specific framework for clinical abbreviation disambiguation that reformulates the challenge as a discriminative selection task rather than an open-ended generation problem. The framework utilizes anchor-guided semantic retrieval and synthetic context prototypes to facilitate a dual-pathway candidate construction process, ensuring terminological precision in clinical contexts.
Who would benefit from the new knowledge in this paper	Researchers in clinical informatics, developers of automated medical coding systems, and healthcare institutions seeking privacy-preserving, locally deployable NLP solutions for standardizing clinical records.

2. Related works

Approaches to clinical abbreviation disambiguation have transitioned from rule-based methods to the current LLMs. While LLMs provide unprecedented contextual understanding, their generative nature often struggles with terminological precision, making the integration of structured sense inventories essential for model constraint. Our study builds on this foundation by refining the relationship between retrieval and selection.

2.1. LLM-based disambiguation of clinical abbreviations

LLMs have recently demonstrated exceptional performance in medical abbreviation disambiguation, driven by Transformer-based context modeling capabilities [31 32]. Studies show that these models can effectively interpret abbreviations by leveraging broad contextual cues [27,33,34]. Specifically, generative models like GPT utilize their semantic understanding to disambiguate abbreviations in multilingual

clinical texts, achieving impressive results even in zero-shot settings without explicit annotations [19,20,35]. Furthermore, their inherent cross-domain and cross-lingual transferability allows for effective adaptation to specialized medical sub-domains through fine-tuning [36,37].

Despite their potential, LLMs in clinical disambiguation often struggle with hallucinations and a lack of terminological standardization, particularly when encountering rare abbreviations [38,39]. While fine-tuning offers improvement, it remains constrained by the scarcity of large-scale, high-quality labeled datasets [40,41]. To address these bottlenecks, recent studies have begun integrating LLMs with structured knowledge sources to enhance reliability [42,43]. Building on this trend, our study combines clinical sense inventories with LLMs to shift the task from open-ended generation to constrained selection. Furthermore, inspired by prompt engineering techniques [35,44,45], we employ customized prompts to generate pseudo-contextual prototypes. This strategy enriches the candidate selection process, enabling the model to function as a discriminative judge rather than an unstable generator.

2.2. Sense inventory for clinical abbreviations

Previous studies have significantly advanced the construction of clinical sense inventories, establishing essential standards for clinical concept normalization. For instance, Liu et al. [46] constructed a medical abbreviation sense inventory using terms from the UMLS. Chang et al. [47] employed logistic regression to compile a biomedical abbreviation dictionary. Additionally, Xu et al. [48] demonstrated that clustering methods could effectively identify senses in clinical corpora, offering a scalable alternative to manual annotation. Moon et al. [49] developed a comprehensive inventory by combining clinical notes and medical dictionaries, covering the most common abbreviations across 352,267 clinical oral notes. Similarly, Grossman et al.'s Metathesaurus [50] integrates eight source lists from various medical specialties, encompassing 104,057 abbreviations and 170,426 expansions. Furthermore, their subsequent work on automated sense inventory generation [51] provides a scalable solution for the dynamic updating of clinical abbreviation libraries.

2.3. Relationship to RAG and multiple-choice QA

S-ACAD shares structural elements with retrieval-augmented generation (RAG) and candidate-selection frameworks, yet its architecture reflects two clinically motivated design choices that distinguish it from both paradigms. First, unlike RAG systems that retrieve using the original query [52], S-ACAD utilizes an intermediate anchor representation generated by the LLM from the surrounding clinical context. This approach is necessary because clinical abbreviations are highly underspecified inputs. The anchor effectively bridges the semantic gap between the abbreviated surface form and the terminology-level sense inventory.

Second, while multiple-choice QA systems select from static options [53], S-ACAD constructs its candidate pool dynamically through two complementary pathways. These consist of phrase-level semantic retrieval in Stage 2 and pseudo-text prototyping in Stage 3. In the latter stage, the synthesized pseudo-clinical records enable a context-level comparison between potential expansions and the observed patient note. This operation achieves semantic complementarity to phrase-level matching.

3. Material and method

Inspired by previous work utilizing BERT-like models for clinical abbreviation disambiguation [54] and LLMs for domain-specific multiple-choice tasks [55], this study proposes a selection-based paradigm, as

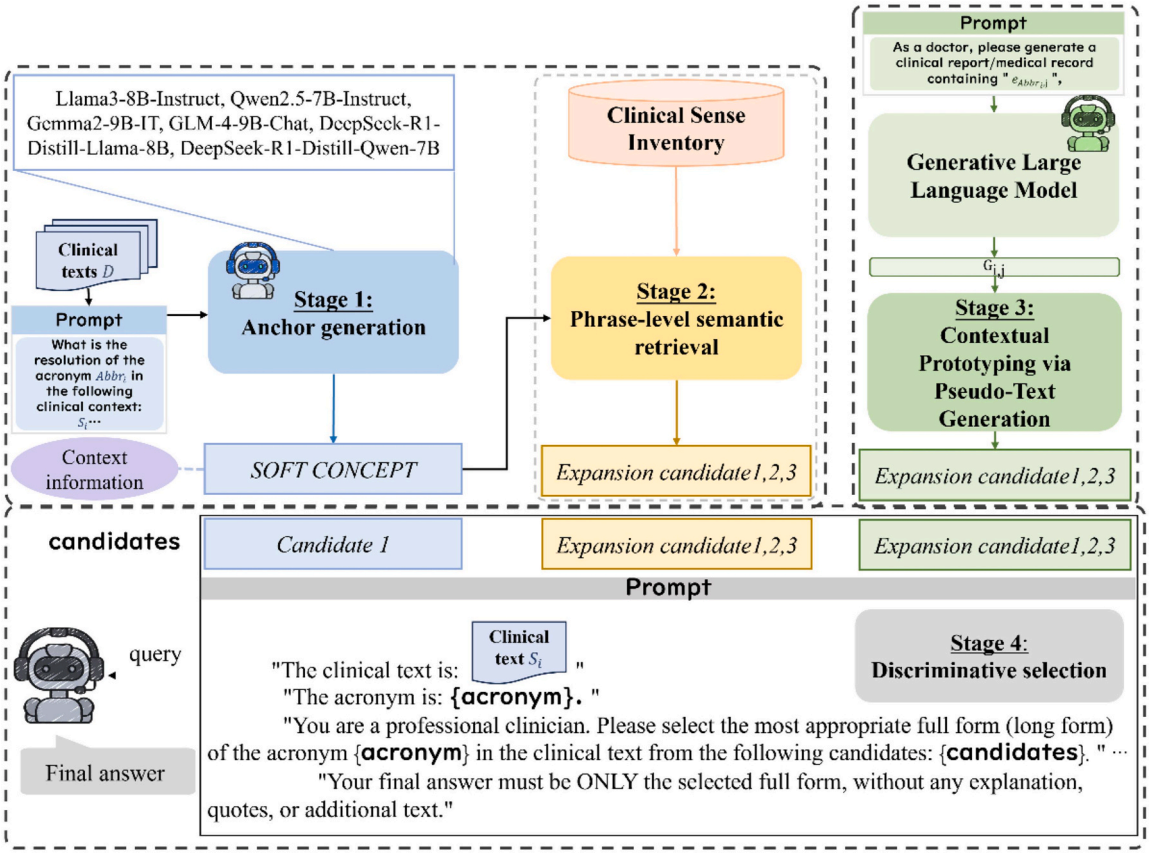


Fig. 1. The architecture of the proposed S-ACAD framework integrating generative LLMs and dual-pathway candidate construction process.

shown in Fig. 1. Unlike approaches that rely solely on direct LLM generation, our method employs a selection-anchored framework. This framework synthesizes generative anchoring (Stage 1) with contextual prototyping via pseudo-text generation (Stage 3), integrating multiple sources of evidence into a single candidate pool for the final model decision. Specifically, **Stage 1: Anchor generation:** Initially, we adopt the predefined prompt template from Kugic et al. (2024) [20] to query the LLM for a preliminary interpretation of the abbreviations $Abbr_i$ within its textual context record S_i from the clinical dataset D. This initial output serves as the generative anchor. **Stage 2: Phrase-level semantic retrieval:** For each $Abbr_i$, we utilize a clinical sense inventory E_{Abbr_i} to compute the semantic similarity between the LLM's initial response $A(S_i)$ and all possible expansions in the inventory. The top-3 most similar expansions are retrieved as knowledge-aligned candidates. **Stage 3: Contextual prototyping via pseudo-text generation:** To further enrich the evidence, the LLM is prompted to generate pseudo-text G_i,j for each expansion in the inventory. We calculate the semantic similarity between the original clinical context S_i and each generated pseudo-text G_i,j. Based on this ranking, we select the expansions corresponding to the top-3 most similar pseudo-texts as context-aware candidates. **Stage 4**: **Discriminative selection**: In the final stage, the initial response from Stage 1 and the six candidates retrieved from Stages 2 and 3 are aggregated into a 7-option candidate pool. The same LLM is then employed as a discriminative judge, selecting the single most accurate expansion from this structured list.

3.1. Anchor generation

Fig. 2 illustrates the anchor generation stage. This stage initiates the disambiguation process by generating a "soft concept" expansion, $A(S_i)$, using the LLM's internal knowledge as a generative anchor. To optimize efficiency within computational constraints, we utilize six open-source LLMs (7B to 9B parameters) to individually process each clinical record S_i ($S_i \in D$) containing an abbreviation $Abbr_i$. This process employs a zero-shot prompt template ($prompt_1$) from Kugic et al. (2024)[20], designed to elicit a concise expansion while explicitly precluding extraneous explanations:

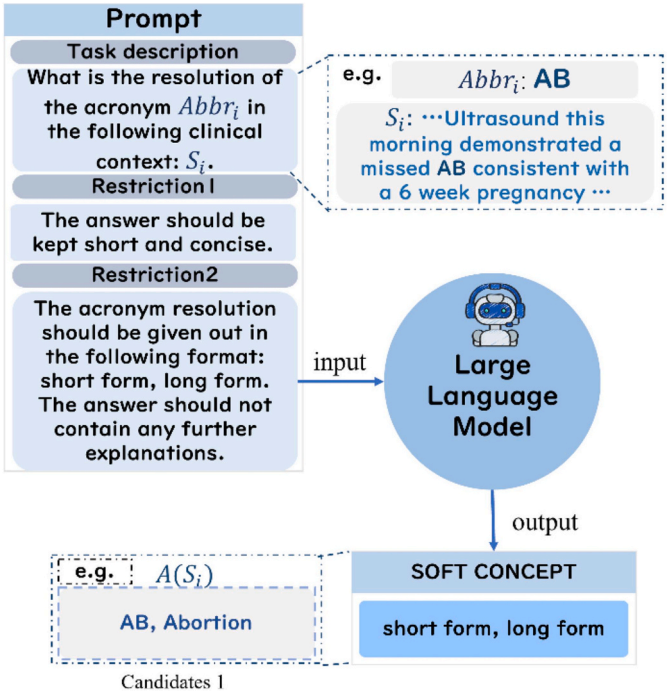


Fig. 2. Schematic diagram of anchor generation stage, exemplified by the abbreviation AB.

$$A(S_i) = M(S_i, Abbr_i, prompt1) \quad (1)$$

Formally, $A(S_i)$ represents the predicted expansion for the i-th record, as generated by the specific LLM instance M conditioned on $prompt_1$.

By instructing the model to identify the short and long forms within the provided clinical context, we establish a preliminary semantic baseline for subsequent stages. The full prompt template is detailed in Table A.1.

3.2. Phrase-level semantic retrieval

In this stage, we utilize a domain-specific sense inventory of clinical abbreviations to perform semantic matching between the "anchor" $A(S_i)$ generated by the LLM and the expansions E_{Abbr_i} in the inventory. This process, as illustrated in Fig. 3, selects the three expansions from the inventory that are most semantically similar to $A(S_i)$, incorporating them as candidates for the final selection stage.

The set of all possible expansions for a given abbreviation $Abbr_i$ within the sense inventory is defined as:

$$E_{Abbr_i} = \{e_{Abbr_i,1}, e_{Abbr_i,2}, \ldots, e_{Abbr_i,N_i}\}$$

where N_i is the total number of distinct senses for $Abbr_i$.

To quantify semantic similarity, we utilize BioSentVec [56] (based on Sent2vec [57]), a sentence embedding model pre-trained on PubMed and MIMIC-III data [58], ensuring high efficacy for clinical text encoding. Both the "soft concept" $A(S_i)$ and each candidate expansion $e_{Abbr_i,j}$ from the sense inventory are converted into vector representations, v_{A_i} and $v_{e_{Abbr_i,j}}$, respectively, using the BioSentVec encoding function $f_{\text{BioSentVec}}(\cdot)$:

$$v_{A_i} = f_{\text{BioSentVec}}(A(S_i)) \quad (2)$$

$$v_{e_{Abbr_i,j}} = f_{\text{BioSentVec}}(e_{Abbr_i,j}) \quad (3)$$

The semantic similarity is then measured by calculating the cosine similarity between these vectors:

$$\langle v_{A_i}, v_{e_{Abbr_i,j}} \rangle = \frac{v_{A_i} \cdot v_{e_{Abbr_i,j}}}{\|v_{A_i}\| \|v_{e_{Abbr_i,j}}\|} \quad (4)$$

Subsequently, the top-3 expansions $E_{Abbr_i}^{\text{top3_sim}}$ exhibiting the highest similarity scores from E_{Abbr_i} are retrieved as the second set of candidates:

$$E_{Abbr_i}^{\text{top3_sim}} = \arg \max_{e_{Abbr_i,j} \in E_{Abbr_i}, \left|E_{Abbr_i}^{\text{top3_sim}}\right|=3} \langle v_{A_i}, v_{e_{Abbr_i,j}} \rangle \quad (5)$$

3.3. Contextual prototyping via pseudo-text generation

In this stage, a generative large language model is prompted to generate pseudo-texts for each expansion $e_{Abbri,j}$ of the abbreviation $Abbr_i$ found in the original clinical record S_i. These generated pseudo-texts are then semantically compared against the original clinical record S_i for candidate selection, as illustrated in Fig. 4.

Specifically, for each expansion $e_{Abbri,j}$, the large language model M is instructed using prompt $prompt_3$ to generate a corresponding pseudo-clinical record, $G_{i,\,j}$, acting as a clinical domain expert:

$$G_i,j = M(e_{Abbri,j}, prompt_3) \quad (6)$$

The detailed prompt template for pseudo-record generation is provided in Table A.1.

Following generation, the BioSentVec model is again employed to encode both the original clinical record S_i and each generated pseudo-record $G_{i,j}$ into vector representations, denoted as v_{S_i} and $v_{G_{i,j}}$, respectively. The cosine similarity, $\langle v_{S_i}, v_{G_{i,j}} \rangle$, is then computed between the two vectors. The top-3 expansions corresponding to the most similar

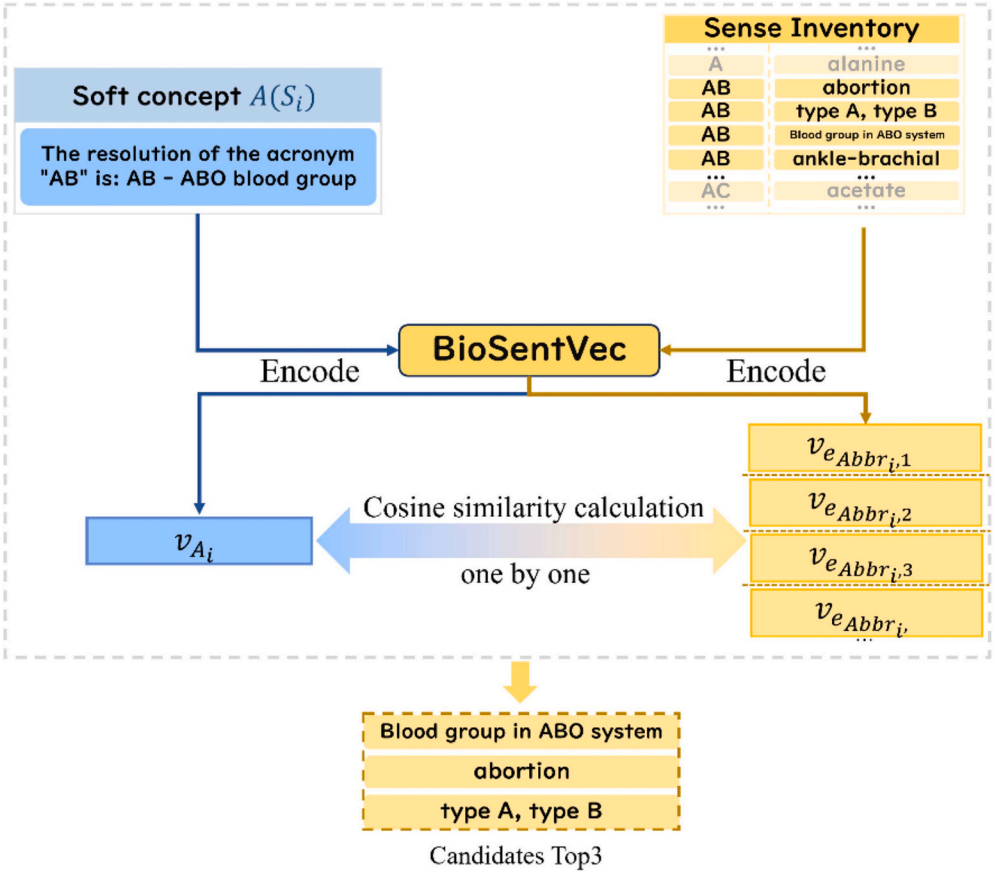


Fig. 3. Schematic diagram of the phrase-level semantic retrieval candidates from clinical sense inventory.

pseudo-texts are selected as the third set of candidates:

$$E_{Abbr_i}^{\text{top3_pseudo}} = \arg \max_{e_{Abbr_i j} \in E_{Abbr_i}, \left|E_{Abbr_i}^{\text{top3_pseudo}}\right|=3} \langle v_{S_i}, v_{G_{i,j}} \rangle \tag{7}$$

3.4. Discriminative selection

The final stage transforms the disambiguation task into a structured multiple-choice question, as shown in Fig. 5, leveraging the LLM's superior discriminative capabilities. We aggregate the candidates from the previous stages into a candidate pool P_i of up to 7 options, where duplicated candidates are removed:

$$P_i = \{\mathrm{A}(S_i)\} \cup E_{Abbr_i}^{\text{top3_sim}} \cup E_{Abbr_i}^{\text{top3_pseudo}} \tag{8}$$

In this stage, the LLM is assigned the role of a professional clinician and guided by a task-specific prompt ($prompt_4$) to select the most appropriate expansion from the candidate pool P_i. Given the original context S_i and target abbreviation $Abbr_i$, the model must interpret the Stage 1 "soft concept" ($\mathrm{A}(S_i)$) and align it with standard medical terminology before comparing it with other candidates. To ensure the output is suitable for automated evaluation, a strict zero-shot constraint is enforced: the model must output only the selected expansion, with no explanations or conversational filler. This prompt focuses the model's discriminative potential entirely on semantic accuracy and standardized normalization. The full selection prompt ($prompt_4$) is detailed in Table A.1.

3.5. Evaluation methods

3.5.1. Evaluation data and metrics

Abbreviation definition

In this work, an "abbreviation" is defined as a shortened form consisting of capital letters derived from the initial letters or syllables of a full expression. This definition encompasses both initialisms and acronyms without distinction.

Datasets.

We evaluated multiple LLMs on two versions of the English CASI dataset: the original full version [49] and the denoised version by Adams [59]. While the original dataset contains noise (e.g., duplicate entries, parsing errors, and identical long/short forms), the denoised version provides a higher-quality benchmark by removing these artifacts. The key characteristics of both datasets are summarized in Table 1. To maintain data integrity, we performed no additional preprocessing or text modifications; specifically, all anonymization tags were preserved prior to prompting.

$$Accuracy = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}(e_{final,i} = e_{gold,i}) \tag{9}$$

Accuracy is employed as the primary metric and calculated as the proportion of records where the model's final selection e_{final} aligns with the ground truth e_{gold} across the total dataset of size N:

where $\mathbb{I}(\cdot)$ is the indicator function.

$$Macro\text{-}F1 = \frac{1}{N_A} \sum_{i=1}^{N} \mathrm{Acc}_i \tag{10}$$

In addition to accuracy, we also report the macro-averaged F1-score (Macro-F1), particularly for the denoised dataset (as each abbreviation in the original dataset contains an equal number of instances), to evaluate the model's performance across potentially imbalanced classes. This ensures that frequent and rare abbreviations are treated with equal importance.

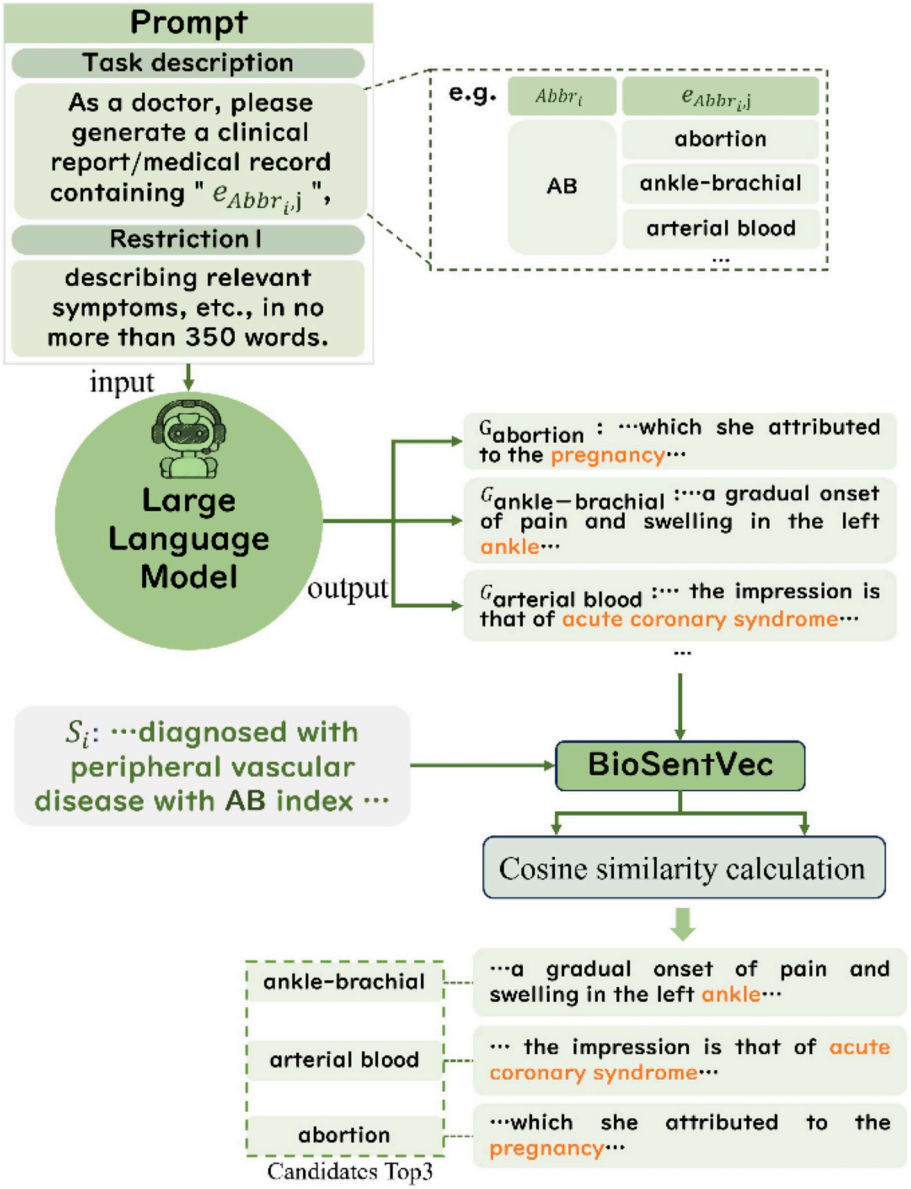


Fig. 4. Schematic diagram of contextual prototyping via pseudo-text generation.

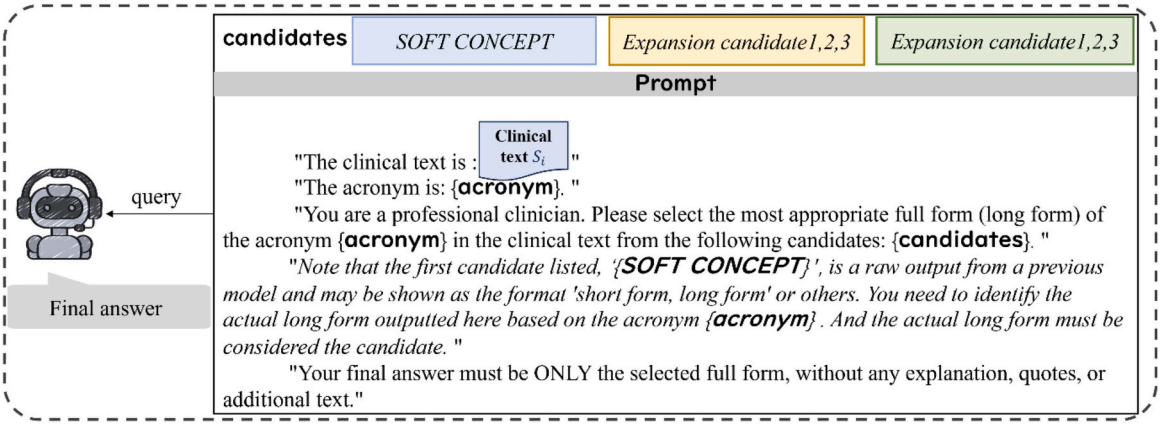


Fig. 5. Schematic diagram of the final discriminative selection.

Here, N_A represents the number of abbreviations, and Acc_i represents the disambiguation accuracy of the i-th abbreviation.

Table 1
Brief information of the original and Adams's denoised CASI.

Dataset	Distinct acronym count	acronym length range	text span count	Mean text span length (words)	pairs
Original CASI	75	[2,4]	37 k	60	352
Adams's Denoised CASI	62	[2,4]	27 k	60	262

3.5.2. LLMs and sense inventory for evaluation

To comprehensively evaluate the robustness of the proposed method, experiments were conducted using six open-source LLMs: LLaMA3-8B-Instruct [60], Qwen2.5-7B-Instruct [61], Gemma-2-9B-IT [62], GLM-4-9B-Chat [63], DeepSeek-R1-Distill-Llama-8B, and DeepSeek-R1-Distill-Qwen-7B. Among them, the latter two are distilled variants of the DeepSeek-R1 inference models [64], based on LLaMA and Qwen architectures, respectively.

Regarding the sense inventory, since most standard inventories, such as UMLS and ADAM, were already integrated into Moon's inventory v2, we utilized Moon's inventory v2 [49] as the source of candidate expansions.

4. Results

4.1. Results of the whole framework

Table 2 presents the comparative performance of six LLMs on Adams's denoised and original CASI datasets. First, regarding model performance, Gemma-2-9B-IT demonstrates superior capability, outperforming all other models by a significant margin. It achieves the highest accuracy of 0.9228 and a Macro-F1 of 0.9279 on the denoised dataset. Following Gemma-2, Llama3-8B-Instruct and Qwen2.5-7B-Instruct exhibit competitive performance, forming a strong second tier with accuracies of 0.8580 and 0.8533, respectively. In contrast, the distilled models (DeepSeek-R1-Distill-Llama and DeepSeek-R1-Distill-Qwen) show relatively lower performance, with accuracies ranging from 0.7107 to 0.7568 on the denoised set, indicating a potential trade-off between model distillation and task-specific accuracy. Critically, the impact of denoising is evident. All models exhibit substantial performance gains on Adams's denoised CASI dataset compared to the original version. For instance, Gemma-2-9B-IT exhibits an absolute accuracy improvement of approximately 0.1643 (from 0.7585 to 0.9228).

Table 2
Accuracy (with 95% confidence intervals) and macro-averaged F1-scores of various large language models on original and Adams's denoised CASI datasets using the S-ACAD framework.

	Adams's Denoised CASI		Original CASI
	Accuracy	Macro-F1	Accuracy(=Macro-F1)
Gemma-2-9B-IT	**0.9228**	**0.9279**	**0.7585**
Llama3-8B-Instruct	0.8580	0.8637	0.6909
Qwen2.5-7B-Instruct	0.8533	0.8610	0.6996
GLM-4-9B-Chat	0.8350	0.8451	0.6889
DeepSeek-R1-Distill-Llama-8B	0.7568	0.7750	0.5961
DeepSeek-R1-Distill-Qwen-7B	0.7107	0.7541	0.5823

* Accuracy and Macro-F1 are equivalent for the original dataset due to its uniform class distribution (equal number of samples per abbreviation).

4.2. Ablation experiment

The complete S-ACAD framework comprises anchor generation (Stage 1), anchor-guided semantic retrieval (Stage 2), contextual prototype construction (Stage 3), and discriminative selection (Stage 4). To rigorously assess the necessity of the anchor mechanism, we designed six targeted ablation configurations. Ablations A and F correspond to Stage 1 alone (Direct generation) and the full pipeline, respectively. Ablation B removes both the anchor and prototype stages, isolating the contribution of contextual semantics provided by these two modules. Ablation C replaces anchor-guided retrieval with raw clinical text retrieval, directly testing whether the anchor provides an irreplaceable semantic bridge in Stage 2. Ablation D bypasses Stages 1 and 2 entirely, evaluating their combined necessity. Ablation E retains anchor-guided retrieval but removes the anchor from the final candidate pool, isolating whether the anchor contributes as a candidate option independent of its retrieval role.

As shown in Table 3, all evaluated LLMs achieve peak performance exclusively when the full pipeline is active, demonstrating the clear superiority of the integrated framework. Taking Gemma-2-9B-IT as a representative case, the full pipeline achieves an accuracy of 0.9228, while the baseline using only Stage 1 direct generation performs substantially lower at 0.6304. Simplified configurations relying solely on abbreviation strings or raw clinical text fail to match the effectiveness of the full pipeline. The most compelling finding is that when the full pipeline is engaged but the anchor is removed from the final selection stage, the accuracy drops from 0.9228 to 0.8643. This confirms that the anchor serves as a critical piece of evidence for the model's final judgment rather than just a retrieval tool.

This pattern remains consistent with the other backbone models. Qwen2.5-7B-Instruct achieves its highest score of 0.8533 under the full configuration, notably outperforming the version without anchor-assisted selection and the setup that bypasses the retrieval stage entirely. Finally, the advantage of the full pipeline is most pronounced in the reasoning-distilled models. Both DeepSeek-R1-Distill-Llama-8B and DeepSeek-R1-Distill-Qwen-7B show substantial improvements in the bottom rows of their respective sections, reaching 0.7568 and 0.7107. These represent marked gains over their initial generative baselines of 0.4121 and 0.2749.

5. Discussion

This study presents S-ACAD, a task-specific framework that transforms clinical abbreviation disambiguation from open-ended generation into a discriminative selection task for LLMs. Our approach demonstrates that providing a selection scaffold using a dual-pathway candidate construction process effectively unlocks the latent medical knowledge within lightweight models. In this section, we analyze the impact of data noise on candidate retrieval, evaluate model performance on context-sensitive polysemy, and compare our approach against existing methods and LLM baselines. Additionally, we provide a systematic error analysis and an assessment of current limitations.

Impact of Data Noise on Candidate Retrieval Performance. To evaluate the effectiveness of our retrieval strategy, we measured the proportion of cases where the correct expansion was successfully included in the candidate pool. Our findings reveal that data noise is the primary bottleneck for retrieval success. On the original noisy CASI dataset, the miss rate attributable to noise exhibited wide variation, frequently resulting in cases where the ground truth was excluded from the candidate pool. This indicates that clinical noise significantly impairs the initial stages of semantic alignment and prototype matching.

Furthermore, we observed varying levels of retrieval robustness among the models. Gemma-2-9B-IT demonstrated the highest stability; even in the presence of noise, it maintained a high coverage rate. In contrast, GLM-4-9B-Chat was highly sensitive, with nearly half of its retrieval failures caused by noisy context. This suggests that the quality

Table 3
Accuracy of various models in ablation experiments on Adams's denoised and original CASI datasets.

LLM	Ablation	Stage1	Stage 2	Stage3	Stage 4	Adams's Denoised CASI	Original CASI
Gemma-2-9B-IT	A	✓	–	–	–	0.6304	0.5969
	B	–	✓*	–	✓	0.8062	0.6317
	C	–	✓**	✓	✓	0.8703	0.7031
	D	–	–	✓	✓	0.8354	0.7491
	E	✓	✓	✓	✓°	0.8643	0.7294
	F	✓	✓	✓	✓	**0.9228**	**0.7585**
Qwen2.5-7B-Instruct	A	✓	–	–	–	0.4929	0.4602
	B	–	✓*	–	✓	0.7674	0.6016
	C	–	✓**	✓	✓	0.8373	0.6691
	D	–	–	✓	✓	0.8184	0.6635
	E	✓	✓	✓	✓°	0.8439	0.6664
	F	✓	✓	✓	✓	**0.8533**	**0.6996**
Llama3-8B-Instruct	A	✓	–	–	–	0.5579	0.5058
	B	–	✓*	–	✓	0.5643	0.5398
	C	–	✓**	✓	✓	0.6793	0.5625
	D	–	–	✓	✓	0.7930	0.6049
	E	✓	✓	✓	✓°	0.6866	0.6651
	F	✓	✓	✓	✓	**0.8580**	**0.6909**
GLM-4-9B-Chat	A	✓	–	–	–	0.5029	0.4574
	B	–	✓*	–	✓	0.6567	0.5174
	C	–	✓**	✓	✓	0.7391	0.5917
	D	–	–	✓	✓	0.7995	0.6412
	E	✓	✓	✓	✓°	0.7397	0.6138
	F	✓	✓	✓	✓	**0.8350**	**0.6889**
DeepSeek-R1-Llama-8B	A	✓	–	–	–	0.4121	0.3495
	B	–	✓*	–	✓	0.4079	0.3833
	C	–	✓**	✓	✓	0.6870	0.5541
	D	–	–	✓	✓	0.7251	0.5287
	E	✓	✓	✓	✓°	0.6932	0.4163
	F	✓	✓	✓	✓	**0.7568**	**0.5961**
DeepSeek-R1-Qwen-7B	A	✓	–	–	–	0.2749	0.2401
	B	–	✓*	–	✓	0.5592	0.5273
	C	–	✓**	✓	✓	0.6299	0.4963
	D	–	–	✓	✓	0.6665	0.5156
	E	✓	✓	✓	✓°	0.6211	0.5417
	F	✓	✓	✓	✓	**0.7107**	**0.5823**

* In the table, "√*" indicates that only the abbreviation is used for semantic retrieval; "√**" indicates that only the clinical text is used for semantic retrieval; and "√°" indicates that the anchor is excluded from the final selection stage.

of the initial "soft concept" generated by the LLM is critical: a robust model like Gemma-2-9B-IT can still generate a semantically relevant anchor from noisy text, ensuring the correct expansion is subsequently retrieved into the candidate pool.

Analysis of Disambiguation Performance and Clinical Context Sensitivity. We further analyzed the specific abbreviation failure cases across different model architectures, providing crucial insights into the interplay between linguistic polysemy and model reasoning. Gemma-2-9B-IT exhibits remarkably high stability on context-safe terms, achieving near-perfect accuracy on AMA and BAL. These results indicate that for terms with fixed clinical patterns, such as "Against Medical Advice" in discharge summaries or "Bronchoalveolar Lavage" in respiratory notes, the model's disambiguation logic is highly robust. In contrast, DeepSeek-R1-Distill-Qwen-7B demonstrates significant volatility when encountering context-sensitive polysemy, with its performance plummeting for acronyms such as OP and MOM. This highlights a critical limitation in resolving terms that span diverse clinical specialties, where the model fails to distinguish between Outpatient and Occiput Posterior, or Milk of Magnesia and Multiple of the Median. Furthermore, the moderate performance on MSSA and NAD reveals a lingering sensitivity to semantic noise, particularly when distinguishing between common physical exam findings and specific biochemical markers.

Error Analysis. We conducted a thorough error analysis on disambiguation failure cases that still exist after being processed through the S-ACAD framework, and summarized the following main reasons:

(1) Inherent high ambiguity and lack of contextual information. For example, abbreviations like "AC" can represent "before meals", "adriamycin cyclophosphamide", or "acetate". When the local contextual information is extremely limited, the model struggles to identify the intended meaning. In such cases, the entire pipeline fails due to insufficient contextual cues.
(2) Transmission and amplification of errors. We observed that incorrect outputs from Stage 1 may misguide subsequent stages. If a semantically irrelevant expansion is predicted in Stage 1, the semantic matcher in Stage 2 tends to rank related but incorrect meanings, propagating the initial error. Additionally, we observed stochastic generative errors. For example, in the stage 4, some outputs from DeepSeek distilled models exhibited repetition loops.
(3) Insufficient reasoning in subtle domain knowledge. In several cases, abbreviations appear in contexts where cues are too subtle or domain-specific for general LLM reasoning. For example, distinguishing "CTA" as "computed tomography angiography" versus "clear to auscultation" usually requires clinical intuition or implicit knowledge of diagnostic procedures, and the information cannot always be directly inferred from the text itself.
(4) Internal limitations in the pseudo-text similarity stage. Even when the correct expansion exists in the candidate pool, the generated pseudo-text may fail to align semantically with the original context, leading to low similarity scores. In addition, in Stage 3, DeepSeek distilled models sometimes misinterpreted the pseudo-text generation task as a sensitive request, leading to refusals.

Comparison with Existing Methods and LLM Baselines. We evaluated S-ACAD against various published methods on the denoised

CASI dataset, with detailed results provided in Table A.2. Our approach shows competitive performance when compared to both large-scale closed-source models and traditional training-based systems.

In the zero-shot category, earlier studies that rely on direct generation have shown clear performance limitations even when using large-scale models. For example, Kugic et al. [20] reported accuracies of 0.8800 with GPT-3.5 and 0.7000 with Llama-2-70b, while Agrawal et al. [27] reached 0.8600 using GPT-3.5. All these methods were tested on Adams's denoised CASI dataset. Unlike these methods that require either expensive APIs or massive parameter scales, our framework uses a moderate-sized open-source model. Notably, it achieved a higher accuracy of 0.9228 and a Macro-F1 score of 0.9279, which represents the best performance currently reported among non-fine-tuned methods. This suggests that reframing the abbreviation disambiguation task from generation to selection may contribute substantially to performance gains, potentially complementing the effect of model scale.

At the same time, although traditional fine-tuned models like BERT combined with BioGPT augmentation [26] and MS_BERT [21] achieve higher accuracy, they depend on heavy clinical annotation and training. S-ACAD effectively bridges the gap between resource-intensive training and lower-precision zero-shot generation. The consistent performance across categories, along with support for local deployment and concept normalization, makes it a practical and scalable choice for real-world clinical environments where data privacy is a priority.

Limitations. Despite its advantages, S-ACAD faces three primary limitations. First, its accuracy is intrinsically tied to the retrieval inventory. Specifically, the system relies on the correct expansion being captured in the candidate pool. Consequently, it may struggle with out-of-vocabulary terms or evolving medical neologisms. Furthermore, although the Moon's inventory incorporates broad resources such as UMLS and ADAM, our evaluation does not establish how S-ACAD behaves under a substantially larger or unconstrained retrieval space. Second, the selection mechanism is sensitive to context quality. In sparse or highly diverse narratives, static prompts may provide insufficient evidence to resolve semantically similar candidates. Future work should explore adaptive prompting to better calibrate model judgment across varied clinical text types. Finally, it is important to note that our evaluation is currently anchored to the CASI benchmark. While CASI is a widely used standard benchmark for clinical abbreviation disambiguation and contains diverse clinical contexts, we recognize that evaluating the framework on a single dataset presents a potential limitation regarding cross-setting robustness. We acknowledge this scope as a constraint of our current experimental design and emphasize that the framework's performance may vary across diverse clinical environments. Expanding this framework to broader clinical datasets, including cross-institutional data and non-English clinical corpora, is a vital direction for future work to further validate the method's generalizability.

6. Conclusion

This study establishes a task-specific framework for automatic clinical abbreviation disambiguation by systematically shifting the task from unconstrained generation to discriminative selection using a dual-pathway candidate construction process for LLMs. We demonstrate that providing a structured candidate pool effectively improves disambiguation accuracy and mitigates generative hallucinations. Our results, particularly the superior performance on Adams's denoised CASI dataset, validate that this selection-anchored strategy represents a potent paradigm for clinical NLP, achieving excellent accuracy even with lightweight models (fewer than 9B parameters). Future research will advance this work in several key directions. First, we will enhance the framework's robustness against complex clinical noise and its adaptability to a diverse range of clinical text types through continuously refined prompt architectures. Second, we will evaluate the system's scalability within unconstrained retrieval spaces and addressing evolving medical neologisms through dynamic knowledge integration. Third, we will extend the method to disambiguate abbreviations in broader clinical datasets, including cross-institutional data and non-English clinical corpora, thereby providing robust support for downstream NLP tasks.

CRediT authorship contribution statement

Siwen Wei: Writing – original draft, Methodology, Data curation. **Xiaoyu Liu:** Writing – review & editing, Validation. **Yonghong Shi:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A1
Prompt Templates.

Stage	Prompt Template
Stage 1 (Preliminary inference)	**What is the resolution of the acronym *Acronym* in the following clinical context: *CONTEXT*. The answer should be kept short and concise. The acronym resolution should be given out in the following format: short form, long form. The answer should not contain any further explanations.**
Stage 3 (Contextual prototyping)	**As a doctor, please generate a clinical report/medical record containing *EXPANSION*, describing relevant symptoms, etc., in no more than 350 words.**
Stage 4 (Discriminative selection**)**	**The clinical text is: *Clinical text*** **The acronym is: *Acronym*.** **You are a professional clinician. Please select the most appropriate full form (long form) of the acronym A*cronym* in the clinical text from the following candidates: *Candidates*.** *Note that the first candidate listed, **SOFT CONCEPT**, is a raw output from a previous model and may be shown as the format 'short form, long form' or others. You need to identify the actual long form outputted here based on the acronym **Acronym**. And the actual long form must be considered the candidate.* **Your final answer must be ONLY the selected full form, without any explanation, quotes, or additional text.**

Table A2
Performance comparison of S-ACAD and prior clinical abbreviation disambiguation methods on the denoised CASI dataset.

Study	Method / Base Model	Paradigm	Accuracy
I. Li et al. [1]	ELMo + Topic	training	0.7041
G. Adams et al[2]	LMC	training	0.7100
A. Kashyap et al. [3]	CLASS-GATOR	training	0.7692
V. Joopudi et al. [4]	CNN	training	0.7783
Hosseini et al. [5]	BERT + BioGPT Augmentation	Fine-tuned	0.9953
Jaber et al. [6]	MS_BERT	Fine-tuned	0.9913
Kugic et al. [7]	GPT-3.5	Zero-shot	0.8800
Agrawal et al. [8]	GPT-3.5	Zero-shot	0.8600
Kugic et al. [7]	Llama-2-70b	Zero-shot	0.7000
Ours (S-ACAD)	**Gemma-2-9B-IT**	**Zero-shot**	**0.9228**

* All zero-shot baselines above follow the preprocessing and denoising guidelines established by Adams et al.

Table A3
Inference efficiency and token consumption across different LLM backbones for S-ACAD.

Backbone	Avg. Time To First Token(TTFT)(s)	Avg. Token Consumption (tokens)
Gemma-2-9b-it	0.5595 ± 0.7212	430.15 ± 33.58
Qwen2.5-7B-Instruct	0.5475 ± 0.7456	478.42 ± 35.01
Llama-3-8B-Instruct	0.5083 ± 0.6885	440.30 ± 35.18
GLM4-9B-chat	0.5843 ± 0.7838	430.04 ± 32.27
DeepSeek-R1-Distill-Llama-8B	0.5381 ± 0.7288	492.14 ± 46.81
DeepSeek-R1-Distill-Qwen-7B	0.5266 ± 0.7172	473.44 ± 44.73

Data availability

Data will be made available on request.

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